

Saumik Saha Kabbya

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Experience

The Data Island [AI Engineer (Part-time), Frontend Engineer (Part-time)] Mar 2025 – Sept 2025
• Developed computer vision and AI-powered solutions for client projects, building backend services with **Python**, **FastAPI**, and **Groq AI**
• Implemented object detection and recognition pipelines using **Ultralytics YOLO** models and **PyTorch**
• Built responsive frontend interfaces with **Next.js** to interface with AI/CV backend systems
• Collaborated remotely with cross-functional teams to deliver end-to-end AI features from model integration to UI

Education

North South University Dhaka, Bangladesh
BS in Computer Science and Engineering Jan 2022 – Dec 2025
GPA: 3.66

Skills

Programming Languages: Python, JavaScript, TypeScript, Java, HTML, CSS
Frameworks & Libraries: React, Next.js, Node.js, Tailwind CSS
AI & ML: Generative AI, Large Language Models (LLMs), LangChain, Retrieval-Augmented Generation (RAG), NLP fundamentals, Computer Vision, Recommendation Systems
Tools: Git, GitHub, Vite, Vercel, Firebase, Supabase, VS Code, Cursor, Antigravity, Qoder

Academic Capstone Project

• **Airis** Jan 2025 - Present
<https://github.com/rajin-khan/Airis>
'Airis', a comprehensive AI-powered vision assistant for the visually impaired using a distributed local-first architecture. The system integrates a wearable **ESP32-CAM** with a local server running **YOLO26s** and **MediaPipe** to provide "Active Guidance" - real-time spatial audio feedback guiding the user to objects with sub-two-second latency. Key features include continuous scene description via **BLIP** and **LLMs**, automated vision-based fall detection with guardian email alerts, and a hands-free **Web Speech API** voice interface, achieving an 87% success rate in object retrieval tasks.

Projects

• **Gravitas** May 2026 - Present
<https://github.com/Kabbya04/Gravitas>
An AI-powered legal document understanding and grounded drafting system featuring **OCR**, **RAG**, **evidence-grounded draft generation**, and **iterative improvement from operator** edits using modern LLM workflows.

• **Hazel Studio** Jan 2026 - Present
<https://www.hazelstudiobd.com/>
A website for the Bangladeshi cosmetics brand Hazel Studio. The whole platform is built on **React** + **Vite** and **Supabase** as the database.

• **Omni Game Recommender System** Apr 2025
<https://github.com/Kabbya04/Omni-Game-Recommender-System>
A content-based game recommendation engine using **Bag of Words**, **TF-IDF**, and **SVM** to suggest similar video games with likelihood predictions.

• **TapSense** Sept 2024 - Dec 2024
<https://github.com/Kabbya04/TapSense>
A cross-platform accessibility application developed with **Flutter** designed for visually impaired users.